1. If the vertices of a triangles are O(0, 0), A (a, 0) and B(0, a). Then, the distance between its circumcenter and orthocenter is:
(a) $\frac{a}{2}$ (b) $\frac{a}{\sqrt{2}}$
(c) $\sqrt{2a}$ (d) $\frac{a}{4}$

MATHS – Coordinate Geometry

2. The straight line x + y = 0, 3x + y – 4 = 0 and x + 3y – 4 = 0 from a triangle which is
(a) Right angled
(b) Equilateral
(c) Isoscales
(d) Isoscales and right angled

MATHS – Straight Line

3. If one of the lines of $ax^2 + 2hxy + by^2 = 0$ bisects the angle between the axes in the first quadrant, then
(a) $h^2 - ab = 0$ (b) $h^2 + ab = 0$
(c) $(a + b)^2 = h^2$ (d) $(a + b)^2 = 4h^2$

MATHS – Pair of Lines

4. Let E be the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and C be the circle$x^2 + y^2 = 9$. Let P and Q be the points (1, 2) and (2, 1) respectively. Then
(a) Q lies inside C but outside E
(b) Q lies outside both C and E
(c) P lies inside both C and E
(d) Plies inside C but outside E

MATH – Conic Section (Ellipse/Circle)

5. A triangle with vertices (4, 0), (-1,-1), (3, 5) is
(a) Isosceles and right angled
(b) Isosceles but not right angled
(c) Right angled but not isosceles
(d) Neither right angled nor isosceles

MATH – Coordinate Geometry (Triangle)

6. Match List I with List II

LIST I		LIST II	
(A)	The angle between the straight lines, $2x^2 + 3y^2 - 7xy = 0$ is	(i)	$\tan^{-1}\frac{3}{5}$
(B)	The circles $x^2 + y^2 + x + y = 0$ and $x^2 + y^2 + x - y = 0$ intersect at angle	(ii)	25π
(C)	The area of circle centred at (1, 2) and passing through (4, 6) is	(iii)	$\pi/4$
(D)	The parabolas $y^2 = 4x$ and $x^2 = 32y$ intersect at point (16, 8) at angle	(iv)	$\pi/2$

Choose the correct answer from the options given below:
(a) A-IV, B-III, C-I, D-II
(b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-II, D-I
(d) A-III, B-IV, C-I, D-II

MATH – Coordinate Geometry

7. If a Chord which is normal to the parabola y^2=4ax at one end subtends a right angle at the vertex, then its slope is -
(a)1 (b)$\sqrt{3}$
(c) $\sqrt{2}$ (d) 4.2

MATH – Parabola (Normal)

8. A straight line has equation $y = -x + 6$ which of the following line is parallel to it?
(a) $2y + 3x = -5$ (b) $-3x - 3y + 7 = 0$
(c) $2y = -x + 12$ (d) $y - x = \frac{1}{10}$

MATH – Straight Line

9. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: If two circles intersect at two points, then the line joining their centres is prependicular to the common chord
Reason R: The perpendicular bisectors of two chords of a circle intersect at its centre.
In the light of the above statements, choose the correct answer from the options given below:
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

MATH – Circle Geometry

10. The tangent to the hyperbola $x^2 - y^2 = 3$ are parallel to the straight line $2x + y + 8 = 0$ at the following points:

(a) (2, 2), (1, 2) (b) $(2, -2), (-2, 1)$

(c) $(-2, -1), (1, 2)$ (d) $(-2, -1), (-1, -2)$

MATH – Hyperbola

11. A circle S passes through the point (0, 1) and is orthogonal to the circles $(x-1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then

(A) Radius of S is 8 (B) Radius of S is 7

(C) Centre of S is (-7, 1) (D) Centre of S is (-8, 1)

(a) A only (b) A and B only

(c) B and C only (d) D only

MATH – Circle

12. Which of the following statement are TRUE?

(A) A equation $ax^2 + bx + c = 0$ has real and distinct roots if $b^2 - 4ac \geq 0$ and $a \neq 0$

(B) The unit digit in 49^{18} is 1

(C) If two lines make complementary angles with the axis of x then the product of their slopes is 1.

(D) The line bx – ay = 0 meet the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Choose the correct answer from the options given below

(a) A and D only (b) B and C only

(c) A, B and C only (d) A, B and D only

Algebra + Coordinate Geometry (Mixed Concept MCQ)

13. Match List-I with List-II

List-I		List-II	
A.	Eccentricity of the conic $X^2 - 4x + 4y + 4y^2 = 12$ is	I.	10/3
B.	Latus rectum of conic $5x^2 + 9y^2 = 45$ is	II.	1
C	The straight line x + y = a touches the curve y = x – x^2 then value of a is	III.	2
D.	Eccentricity of conic $3x^2 - y^2 = 4$ is	IV.	$\sqrt{3/2}$

Choose the correct answer from the options given below

(a)A-I, B-II, C-IV, D, III (b) A-II, B-I, C-III, D-IV

(c) A-IV, B-I, C-II, D-III (d) A-IV, B-II, C-I, D-III

Conic Sections – Ellipse & Hyperbola Properties

14. A line passes through a point (2,3) such that sum of its intercepts on the axes is 12 then equation of line (/s) is (/are) given by

(A) 3x + y =9 (B) x + 3y = 9

(C) x+ 2y =8 (D) 5x + 7y = 35

Choose the correct answer from the options given below

(a) A only (b) A, B and C only

(c) A and C only (d) B, C and D only

Straight Lines – Intercept Form

15. The equation of a circle that passes through the points (3,0) and (0,–2) and its centre lies on a line 2x+3y = 3 then equation of the circle is given by

(a) $x^2 + y^2 + 2x + 16y + 72 = 0$

(b) $10x^2 + 10y^2 - 6x - 16y - 72 = 0$

(c) $5x^2 + 5y^2 + 6x + 16y + 72 = 0$

(d) $10x^2 + 10y^2 + 6x + 16y - 72 = 0$

Circle – Equation of Circle

16. If the vertices of a triangle are (1,2), (2,5) and (4,3) then the area of the triangle is

(a) 3 sq. units (b) 4 sq. units

(c) 6 sq. units (d) 8 sq. units

Coordinate Geometry – Area of Triangle

17. An equilateral triangle is inscribed in a parabola $y^2 = 8x$ whose one vertex is at the vertex of the parabola then the length of the side of the triangle is:

(a) $8\sqrt{3}$ units (b) $16\sqrt{3}$ units

(c) $4\sqrt{3}$ units (d) $\sqrt{3}/2$ units

Parabola – Geometry Application

18. A equation of a conic is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, where a, b, c, f, g and h are contants Then which of the following statement are true

(A) The given conic is circle if a = 0 and b = 0

(B) The given conic is circle if a = b ≠ 0 and h = 0

(C) The given conic is circle if a = b ≠ 0 and h≠ 0

(D) The given conic represents a pair of real and distincts straight lines if f = g = c = 0 and $h^2 - ab > 0$

Choose the correct answer from the options given below

(a) B only (b) B and D only

(c) A, B C and D (d) D only

Conic Sections – General Second Degree Equation

19. The points $(K, 2-2K), (-K+1, 2K)$ and $(-4-k, 6-2K)$ are collinear if:

(CUET-PG 2025) Co-ordinate

(A) $K = \frac{1}{2}$ (B) $K = \frac{-1}{2}$
(C) $K = \frac{3}{2}$ (D) $K = -1$
(E) $K = 1$

Choose the **correct** answer from the options given below:

(a) (A) and (D) only (b) (A) and (E) only
(c) (B) and (D) only (d) (D) only

20. The length of major axis and coordinate of vertices for the ellipse $3x^2 + 2y^2 = 6$ respectively are:

(CUET-PG 2025) Ellipse

(a) $2\sqrt{2}$, $(0, \pm\sqrt{3})$ (b) $2\sqrt{3}$, $(0. \pm\sqrt{3})$
(c) $2\sqrt{2}$, $(\pm\sqrt{3}, 0)$ (d) $2\sqrt{3}$, $(\pm\sqrt{3}, 0)$

21. If $x^2 = -16y$ is an equation of parabola then:
(A) directrix is $y = 4$
(B) directrix is $x = 4$
(C) co-ordinates of focus are $(0,-4)$
(D) co-ordinates of focus are $(-4,-0)$
(E) length of latusrectum = 16

(CUET-PG 2025) Parabola

Choose the correct answer from the options given below:
(a) (A) and (E) only
(b) (B), (C) and (E) only
(c) (A), (C) and (E) only
(d) (B), (D) and (E) only

22. The center and radius for the circle $x^2 + y^2 + 6x - 4y + 4 = 0$ respectively are:

(CUET-PG 2025) Circle

(a) (2, 3) and 3
(b) (3, 2) and 8
(c) (2, -3) and 3
(d) (-3, 2) and 3

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
b	c	d	d	a	c	c	b	d	d
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
c	b	c	c	b	b	b	b	a	b
21.	22.								
c	d								